
TE Computer (V) Sub: Theory of Computation Date: 19th Aug 2025

Assignment No 02

Topic: Grammars

CO3	Design, identify and simplify Context free grammar, apply the concepts of grammar for context free languages.
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1. Define and explain
 - a. Context Free Grammar
 - b. Regular Grammar
 - c. Context Free Language
 - d. Chomsky hierarchy of grammar
 - e. Define closure properties of CFL
 - f. Pumping Lemma for CFL with example

2. Consider following CFG. Is it already simplified? Explain you answer. Convert it to CNF form.

$$S \rightarrow ASB \mid a \mid bb$$
$$A \rightarrow aSA \mid a$$
$$B \rightarrow SbS \mid bb$$

3. Illustrate the nature of all 4 types of grammars with its production rules
4. Prove the equivalence of left linear and right linear regular grammar.

5. The grammar G is

$$S \rightarrow aB \mid bA$$
$$A \rightarrow a \mid aS \mid bAA$$
$$B \rightarrow b \mid bS \mid aBB$$

Derive using Left Most Derivation (LMD) and Rightmost Derivation (RMD) for the following string "aaabbb". Draw Parse Tree. Check whether the grammar is ambiguous or not.

6. Consider the grammar

$$S \rightarrow 0B \mid 1A$$
$$A \rightarrow 0 \mid 0S \mid 1AA$$
$$B \rightarrow 1 \mid 1S \mid 0BB$$

For the string, 00110101 find the leftmost derivation, rightmost derivation and parse tree. Check whether the grammar is ambiguous or not.

- 7.

Consider the following grammar $G = \{ V, T, P, S \}$, $V = \{ S, X \}$, $T = \{ a, b \}$ and productions P are: $S \rightarrow aSb \mid aX$

$X \rightarrow Xa \mid Sa \mid a$

Convert the grammar in Greibach Normal Form.

8. Construct CFG to generate the language

$$L = \{ a^i b^j c^k \mid k=i+j, i, j \geq 1 \}$$

Find the Context Free Grammar for following

9. i. $L = \{ a^i b^j c^k \mid i = j+k \}$ ii. $L = \{ x \in \{0,1\}^* \mid x \text{ has equal number of zeros and ones} \}$

10. Define pumping Lemma for Context Free Languages. Prove with pumping lemma that the language $L_{012} = \{ 0^n 1^n 2^n \mid n > 0 \}$ is not CFL.